

Ceremonial Entheogen Use in Puerto Rico: A Descriptive Survey of Participants and a Small Facilitator Subsample

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Abstract

Ceremonial entheogen use occurs in Puerto Rico outside clinical and regulatory structures, and the practices surrounding it have not been described. We conducted an anonymous, bilingual, cross-sectional online survey of adults aged 21 and over reporting involvement in entheogen-assisted ceremonies in Puerto Rico, recruited by convenience and chain referral, February–

June 2026. Of 100 submissions, 73 identified a role: 72 as participants and 8 as facilitators, 7 of whom were both. Respondents were middle-aged (mean 42.4 years), mostly university-educated, and almost all resident in Puerto Rico. Most recent ceremonies were typically single, multi-hour events in natural settings involving psilocybin mushrooms or ayahuasca, and satisfaction was strongly ceiling-weighted (median 6 of 6). Preparation was near-universal among those answering while pre-ceremony screening was not, and the small facilitator subsample described training and emergency planning as informal rather than credentialed. Five of 67 participants answering reported ever experiencing non-consensual physical contact during a ceremony, and 2 of 65 at their most recent one. A comparison of any preparation against none, specified in the analysis plan before testing, was positive in direction but rests on five unprepared respondents and is exploratory. These findings describe this sample only and support no prevalence or causal claim.

Keywords: entheogens; psychedelics; ceremonial use; Puerto Rico; harm reduction; survey research

1 Introduction

Ceremonial use of entheogens—psilocybin-containing mushrooms, ayahuasca, and related preparations taken in structured group or guided settings—has become substantially more visible across the Americas over the past two decades, largely outside the clinical-trial infrastructure that generates most of the contemporary research literature (Golden et al., 2022; Carvalho et al., 2025). The resulting evidence base is asymmetric. Randomized clinical research characterizes carefully screened participants receiving standardized doses under monitored conditions, and its samples remain narrow (Hughes and Garcia-Romeu, 2024; Marcus and Rush, 2026), while community ceremonial practice is documented mainly through ethnographic and qualitative work, with comparatively few survey-level descriptions of who takes part and under what conditions (Carvalho et al., 2025).

Puerto Rico occupies an unusual position in this landscape. It is a Spanish-speaking Caribbean

jurisdiction in which psilocybin, DMT, and related compounds are Schedule I substances under the federal Controlled Substances Act (United States Congress, 1970) and the territory’s own aligned statute (Legislative Assembly of Puerto Rico, 1971), and which sits outside the state-level decriminalization and regulated-access reforms reshaping the legal environment in parts of the United States (Siegel et al., 2023). It is also a setting with a long-documented tradition of spiritually framed healing practice—*espiritismo* and related practices—described in the clinical and public health literature for more than four decades (Koss, 1980; Comas-Díaz, 1981; Bird and Canino, 1981; Zerrate et al., 2022).

Local survey evidence on entheogen use is nonetheless minimal. One published Puerto Rico-specific survey exists, conducted by members of the present research team, but it is narrower in scope: it characterizes psilocybin use patterns, motivations, and personality correlates rather than ceremonial practice, and addresses neither facilitation nor safety (Vélez Rodríguez et al., 2026). An unpublished thesis has examined university-community perceptions of psilocybin mushrooms (Doelter Morales and Martínez Miranda, 2020). Neither describes how ceremonies are conducted. Basic descriptive questions—who takes part, which substances are used, where and for how long ceremonies occur, whether participants are prepared or screened beforehand, and what adverse experiences are reported—therefore have no published local answer.

A descriptive characterization contributes in three ways. First, it establishes whether practices reported in a community sample resemble the harm-reduction structures clinical and retreat-based frameworks assume: preparation, pre-ceremony screening, and a plan for difficult experiences (Gorman et al., 2021; Pilecki et al., 2021; Dutton et al., 2025). Second, it identifies which safety outcomes—including boundary violations, shown elsewhere to be measurable in naturalistic samples (Kruger et al., 2025; Peluso et al., 2020)—occur at frequencies justifying measurement in a larger study. Third, it produces the instrument-level evidence needed before any confirmatory study is designed here. None of this depends on a test reaching significance: the description and its measurement lessons are the result.

Our objectives were to describe the sample and the ceremonial context, and to examine bivari-

ate associations between preparation and reported ceremony outcomes and between participant-observable facilitator practices and perceived safety. A safety and adverse-event domain, absent from the original protocol, was added during analysis. Every objective is descriptive. The design supports no statement about how common these practices are in Puerto Rico, and no statement about what any practice produces.

2 Methods

2.1 Design, ethics, and consent

This was a single-timepoint, cross-sectional, self-administered online survey of adults reporting involvement in entheogen-assisted spiritual ceremonies in Puerto Rico. The sample was non-probability and the analysis unweighted. The study was approved by the Institutional Review Board of the University of Puerto Rico, Medical Sciences Campus (protocol 2509472552, approved 4 November 2025) and was fielded on KoboToolbox between 24 February and 9 June 2026. Participation was fully anonymous: no names, dates of birth, or other personal identifiers were collected. The information sheet—covering purpose, procedures, risks, voluntariness, and data retention—was presented within the questionnaire itself, and respondents who read it and proceeded to the items thereby gave informed consent. This procedure was described in the protocol submitted for ethical review and approved as described, so no signed consent document was collected or required. Every item offered “prefer not to answer”, withdrawal was permitted at any point, no compensation was offered, and mental-health resources were listed. Reporting follows STROBE (checklist supplied as Supplementary Material).

2.2 Recruitment and sample construction

Recruitment used convenience sampling with chain referral, promoted through Facebook, Instagram, Twitter/X, and WhatsApp and by bulletin-board announcements at the University of Puerto

Rico Río Piedras and Medical Sciences campuses. Inclusion criteria were age 21 or older, ability to read and write in Spanish, access to an internet-connected device, and prior participation in an entheogen-assisted spiritual ceremony in Puerto Rico. These criteria were self-assessed and advisory rather than enforced by instrument logic: only age gated the questionnaire, the form was fielded bilingually, respondents reporting no ceremony involvement reached the role fork regardless, and residence in Puerto Rico was never required. The recruitment target of 100 responses was met.

Of the 100 submissions collected, 13 were abandoned before any substantive item and excluded, leaving 87 respondents who reached the role-identification questions. Of those, 14 answered “no” to both role items, so 16% of those reaching the role fork screened out; they are not described as a subgroup. The analytic sample is the remaining $N = 73$ respondents who identified a role. Roles are not a partition: 72 identified as ceremony participants and 8 as facilitators, 7 of the 8 facilitators also identifying as participants, so the facilitator branch is largely a subset of the participant branch rather than a separate group of people. The branches are nonetheless *unlinked at the ceremony level*—the instrument carries no ceremony or facilitator identifier, and for a dual-role respondent the ceremonies led are different events from those attended—so facilitator- and participant-reported measures cannot be crossed. One further routing rule governs every participant-side base: dual-role respondents reached the participant block only after a continuation prompt, and 3 of the 7 answered “no” and were shown no participant item. The participant branch is therefore 72 self-identified respondents of whom 69 were routed, those 3 appearing as structural skips throughout (Figure 1).

2.3 Instrument

No validated Puerto Rico-specific instrument for characterizing ceremonial entheogen use exists, so the questionnaire was developed by the research team, reviewed iteratively, professionally translated, and given a final review for cross-language consistency. It was deployed as one bilingual XLSForm with Spanish as the default and English available through the platform’s language

switcher, so open-text responses were mixed-language.

The primary outcome, satisfaction with the most recent ceremony, asked “From 0 to 6, how pleasant was the ceremony?” with anchors from *very unpleasant* (0) to *very pleasant* (6), and was treated as ordinal. The primary exposure asked whether, aside from the facilitator’s instructions, the respondent personally engages in preparation for these ceremonies (yes / sometimes / no), capturing self-directed preparation only. The two differ in referent: the outcome asks about one event and the exposure about a general practice, so the primary analysis relates a single-ceremony outcome to a habitual exposure. Secondary participant-side measures—in-ceremony comfort and safety, trust in the facilitator, prior knowledge of the facilitator’s background, whether screening questions were asked, and prior disclosure to a health professional—all refer to the most recent ceremony. The non-consensual contact item asked whether the respondent had *ever* experienced physical touch during a ceremony that felt unexpected or that they had not clearly agreed to beforehand, explicitly including moments of uncertainty or absent recall; its breadth exceeds a narrower assault framing and the resulting figure should be read accordingly. A parallel item asked the same of the most recent ceremony alone, and both are reported. That lifetime referent does not match the facilitator-practice measures crossed against it in §3.4, a mismatch restated where those results appear. Facilitator-branch items covered training, screening approach, distress-response protocol, emergency planning, and crisis signs witnessed. Three substance variables were kept distinct and never merged: facilitator-led substances, participant substances across all ceremonies attended, and the substance used in the most recent ceremony. Attention checks were branch-specific. Verbatim wording and anchors appear in Supplementary Table S1.

2.4 Missing data, structural skips, and disclosure

Denominators shift across items for two distinct reasons, kept separate throughout: structural skips arise from branch logic and are reported as their own row in Tables 2 and 3, while item nonresponse is the residual. Every figure carries its own base. “Prefer not to answer” and “I’m not sure” were retained as substantive categories in all frequency tables, excluded from ordinal numeric tests

because they carry no rank position, and treated as item-missing rather than scored as zero in composite indices.

Counts are reported exactly throughout, including cells below five. The disclosure protection this study relies on is its design rather than cell masking: no direct identifiers were collected, all reporting is aggregate, and no free-text response is quoted anywhere. A masking convention would have been cosmetic in any case, because every table states the base against which its percentages are computed, from which any masked count is recoverable.

Because the platform populated no duplicate-detection metadata and recorded no IP or device information, duplicate submissions were screened manually on the respondent-chosen nickname and password; no analytic row required exclusion on duplicate grounds, and the two respondents who failed an attention check were flagged rather than dropped, per protocol (Supplementary Material §S7).

2.5 Statistical approach

The outcome distribution was strongly ceiling-weighted and several exposures had thin or unbalanced categories, so all tests were nonparametric: Mann–Whitney U , Kruskal–Wallis H , permutation χ^2 with 10,000 label permutations (seed 20260715) for thin contingency tables, and Fisher’s exact test at very small n . Effect sizes were rank-biserial r_{rb} with the common-language effect size, epsilon-squared ε^2 , and Cramér’s V respectively, with Wilson 95% confidence intervals for prevalences.

The preparation-to-satisfaction comparison was designated in the analysis plan as the primary comparison before any test was run, and is reported uncorrected, with a tie-aware permutation test as a robustness check because the outcome is heavily tied. The study was not pre-registered, and the analysis plan was written after data collection. All other bivariate tests were assigned to four prespecified families—bivariate, substance-stratified, safety-domain, and legal-knowledge—corrected jointly within family by the Benjamini–Hochberg procedure and kept separate because they share neither exposures nor outcome domains (Supplementary Tables S4–S7).

One exploratory proportional-odds model was fitted (Supplementary Table S8). Analyses were conducted in Python; the full pipeline is available (see Data availability).

2.6 Protocol deviations

Three deviations are reported transparently. The protocol’s four-arm preparation $\times$ integration design is not estimable from the collected data: the instrument contains no clean participant integration-received item, and the preparation split is badly unbalanced. Preparation was therefore examined against the outcome directly, and no integration measure is reconstructed anywhere in this paper. The protocol also specified crossing facilitator training and protocol items against participant-perceived safety, and a composite harm-reduction score mixing the two branches; because facilitator self-report cannot be linked to the ceremony any particular participant attended, both would have required pairing unrelated events. Two participant-answerable proxies were substituted—prior knowledge of the facilitator’s background and whether screening questions were asked—facilitator self-report is presented separately and descriptively, and two branch-specific composites replaced the mixed one. Two skip-logic errors in the instrument were also identified (Supplementary Material §S6).

3 Results

3.1 Respondents

The analytic sample of 73 adults skewed middle-aged and highly educated (Table 1). Among the 72 with a valid age, the mean was 42.4 years and the median 41 (SD 11.9, range 21–71, IQR 34–50); 35 of the 71 answering the gender item (49%) identified as women; 56 of the 72 answering the education item (78%) held a bachelor’s degree or above; and 44 (61%) described their economic situation as comfortable against 6 (8%) as difficult. Sixty-eight (94% of the 72 answering) reported residing in Puerto Rico; residency was a descriptive covariate rather than an inclusion filter, and

the presence of respondents living elsewhere tempers any reading of the sample as describing Puerto Rico specifically. Free-text engagement was substantial relative to sample size: 37 open-text fields yielded 285 non-empty answers, roughly three-quarters in Spanish. A coding scaffold for the research team’s thematic analysis accompanies these data (Supplementary Material §S8); the corpus is not interpreted here.

3.2 The ceremonial context

Participants reported extensive but not recent ceremonial histories. Among the 67 answering, the median number of lifetime ceremonies attended was 6 (IQR 3–12, range 1–250) against a median of 1 in the past 12 months (IQR 0–2), and 30 of the 66 answering the recency item (45%) reported their last ceremony more than 12 months earlier.

Across all ceremonies attended, the substances most often reported by the 67 participants answering were magic mushrooms (50, 75%), ayahuasca (43, 64%), tobacco (38, 57%), and cacao (23, 34%), followed by an “other” category (18, 27%) that only 3 respondents specified in its free-text follow-up, leaving it uninterpretable, and then DMT (17, 25%), LSD (13, 19%), MDMA (10, 15%), and ketamine (4, 6%) (Figure 2). For the most recent ceremony, the individually reportable groups were magic mushrooms ($n = 31$) and ayahuasca ($n = 23$), every remaining substance being pooled into a combined group ($n = 11$) under a prespecified rule.

The most recent ceremony was typically a discrete, multi-hour, outdoor event: a single ceremony rather than a retreat for 48 of the 65 answering (74%), a natural setting for 37 of 67 (55%), and most often either 4–6 hours or an overnight event of roughly 12 hours or more (Table 2).

3.3 Preparation, screening, and prior knowledge

Preparation was near-universal among those answering (Table 3): of 67, 55 (82%) prepared and 6 (9%) sometimes did, against 6 (9%) reporting none. This imbalance is why the four-arm design was abandoned and it constrains every preparation analysis below. Participant-observable facilitator practices were common but not uniform: 54 of 67 (81%) knew about the facilitator’s

background beforehand, but only 39 of the 66 answering the screening item (59%) reported being asked screening questions, against 15 (23%) who were not and 7 (11%) who were unsure. Disclosure outside the ceremonial context was uncommon: 8 of the 65 answering (12%) had spoken to a health professional about their participation beforehand.

A participant-side index counting these three practices (range 0–3) was constructed for the 54 with complete data on all three: 35 (65%) reported all three and 16 (30%) two, and *none* reported zero. Internal consistency was low ($\alpha = 0.14$, 95% CI -0.36 – 0.47), consistent both with the items capturing distinct protective practices rather than one latent construct and with their differing referents, since preparation is asked as a general practice while the other two ask about the most recent ceremony. The score is read as a count of practices present rather than as a validated scale.

3.4 Experience and perceived safety

Reported satisfaction was overwhelmingly positive. Of the 72 respondents identifying as participants, 64 answered the satisfaction item—63 giving a numeric score and 1 declining—while 3 were never routed to the participant section and 5 did not respond. The distribution was strongly ceiling-weighted with a median of 6 and the mode at the scale maximum (Figure 3). Perceived in-ceremony comfort and safety was similarly high and undifferentiated, returning a median of 3 in every subgroup tested and differing neither by prior knowledge of the facilitator’s background nor by whether screening questions were asked. Trust in the facilitator was the one participant-side comparison differing at the uncorrected level—median 4 (IQR 3–4) among the 54 who knew the facilitator’s background beforehand against 3 among both smaller groups ($H = 8.15$, $p = 0.017$, $\varepsilon^2 = 0.10$)—and did not survive correction ($q = 0.102$). Satisfaction did not differ significantly by most-recent-ceremony substance ($H = 4.98$, $p = 0.083$, $q = 0.166$). Full test tables are in Supplementary Tables S4–S7.

The safety domain produced the sample’s most consequential descriptive finding. Of the 67 participants answering the lifetime non-consensual contact item, 61 (91%) reported no such contact, 5 (7%) reported that it occurred, and 1 was unsure; the within-sample proportion was 7.5%

(Wilson 95% CI 3.2–16.3). The parallel most-recent-ceremony item, answered by 65, returned 2 affirmative and 62 negative, with 1 declining: of the 5 reporting the experience ever, 2 placed it at the most recent ceremony and 3 did not. That bears directly on what follows, because the facilitator-practice measures crossed against the lifetime item all describe the most recent ceremony, so for those 3 respondents exposure and outcome refer to different events. Non-consensual contact was associated with prior knowledge of the facilitator’s background at the uncorrected level (permutation $\chi^2 = 18.90$, $p = 0.047$, $V = 0.38$, $n = 67$) but not with screening (permutation $\chi^2 = 13.64$, $p = 0.094$, $V = 0.32$, $n = 66$); neither survived correction (both $q = 0.236$), nor did the comparison by substance ($p = 0.486$) or by the harm-reduction index ($p = 0.641$). The research team elected to publish these cross-tabulations rather than withhold them: withholding would remove the only evidence this study can offer on whether such reports cluster with any measured practice, and aggregate reporting of an anonymous survey that collected no identifiers leaves no respondent identifiable. Every result here is hypothesis-generating, rests on the referent mismatch just described, and is powered neither to confirm nor to rule out an association. One measurement observation belongs alongside them: an item on perceived positive life impact had zero variance, all 67 respondents answering selecting the same option.

3.5 Facilitator practice

Eight respondents identified as facilitators, and 7 of the 8 also identified as participants. Every facilitator who answered a substantive facilitator item is one of those 7, so this section and the participant results above describe overlapping people rather than independent samples. Cells are small and no comparison at this base is meaningful. Of the 7 answering, 6 gave personal experience and self-study as their preparation for the role and *none* reported formal certification or academic coursework; on emergency planning, 4 described an informal or unwritten approach and 3 said it depends on the situation, with none selecting a formal written plan. All 7 answered the crisis-signs item: disorientation, feeling overwhelmed, fainting, difficulty breathing, and panic attacks were each reported by 3, and signs of self-harm or suicidal thoughts by 2. Full distributions, including

screening and distress-response practices, and the undefined odds ratios this branch returns, are in Supplementary Tables S2–S3.

3.6 Preparation and reported satisfaction

Among the 63 participants giving a numeric satisfaction score, 58 reported some preparation (median 6, IQR 5–6) and 5 reported none (median 4, IQR 3–5; Figure 4); a sixth respondent reporting no preparation did not answer the outcome, which is why this base is 5 rather than the 6 in Table 3. The Mann–Whitney comparison specified in the analysis plan returned $U = 218.0$, $p = 0.041$, $r_{rb} = 0.50$, and a common-language effect size of 0.75; a tie-aware permutation test gave $p = 0.024$. The structural weakness of this comparison precedes its result: the no-preparation group contains five respondents. The prespecified three-level sensitivity analysis was non-significant ($H = 4.24$, $p = 0.120$, $q = 0.359$), the practice index was unassociated with satisfaction ($H = 2.09$, $p = 0.352$, $q = 0.422$), and an exploratory proportional-odds model adjusting for age and facilitator trust placed preparation short of conventional significance (OR 5.21, 95% CI 0.84–32.41, $p = 0.077$) while trust reached it (OR 2.20, 95% CI 1.06–4.53, $p = 0.033$), at roughly 12 cases per estimated parameter (Supplementary Table S8). No test in any corrected family was significant at $q < 0.05$.

4 Discussion

4.1 A structurally informal practice landscape

The most stable picture this sample supports is a practice organized around informal, relationally transmitted expertise rather than credentialed or documented protocol. Among the facilitators who self-reported, training came from personal experience, self-study, mentorship, and family tradition; none reported formal certification or academic coursework, and emergency planning was uniformly informal or conditional. This sits at some distance from the structured preparation,

screening, and difficult-experience protocols that harm-reduction frameworks for psychedelic settings assume (Gorman et al., 2021; Pilecki et al., 2021; Dutton et al., 2025), is consistent with survey evidence that people who use psychedelics support clearer practice standards (Kruger et al., 2023), and is a reminder that in community settings the containing structure is the setting itself rather than a protocol (Hartogsohn, 2017).

On the participant side the informality reads differently. Preparation was near-universal among those answering, but roughly a quarter reported that no screening questions were asked and a further group was uncertain. The asymmetry is worth separating: preparation is something participants largely do, whether self-directed or instructed, whereas screening must be done *to* them, and was substantially less consistent here. If that asymmetry holds in a larger sample it would locate the intervention target on the facilitator side—but this sample cannot establish whether it holds, and eight facilitators cannot characterize screening practice.

The demographic profile—middle-aged, university-educated, economically comfortable—resembles that of ceremony-attender cohorts described elsewhere, though at the older and more educated end of them. Its mean age of 42.4 years and 78% holding a bachelor’s degree or above compare with 37.6 years and 49% degree-holding among United States neo-shamanic ayahuasca ceremony attenders (Pagni et al., 2025), 37.0 years among ceremonial ayahuasca retreat attenders in the Peruvian Amazon (Ruffell et al., 2021), and 40 years with 59% degree-holding in a large international naturalistic psilocybin cohort (Nayak et al., 2023); the 49% here identifying as women falls inside the 44–61% those three report, as it does for international psychedelic-survey samples (Kopra et al., 2023; Robinson et al., 2026) and this team’s psilocybin-specific Puerto Rico survey (Vélez Rodríguez et al., 2026). That convergence is best read as evidence about who responds to online psychedelic surveys, not about who attends ceremonies (Bethlehem, 2010).

4.2 A ceiling that limits what satisfaction can reveal

Reported satisfaction clustered at the top of its scale, and a perceived-life-impact item had no variance at all. This is a substantive finding—these respondents overwhelmingly described their most

recent ceremony positively, consistent with naturalistic cohorts reporting predominantly positive appraisals (Nayak et al., 2023; Ruffell et al., 2021)—but it also constrains the analysis severely. An outcome with almost no variance in the lower range cannot discriminate between groups, and the null results across the corrected families are consistent with that constraint rather than with the absence of any association. Ceiling behavior is a known property of self-report scales for hallucinogen experience rather than a peculiarity of this instrument (Bouso et al., 2016). Satisfaction, measured this way, is close to exhausted as an outcome here; future work needs measures with headroom in both directions before group comparisons on experience quality can be informative.

4.3 Safety signals the design can identify but not quantify

Five of 67 participants answering reported non-consensual physical contact at some point, two of them at their most recent ceremony, and facilitators reported having witnessed crisis signs at a base of seven at which no rate is estimable. These come from different reporters—none of the five reporting non-consensual contact is a facilitator—and cannot be combined. That such reports appear at all is consistent with survey-based evidence on boundary violations in naturalistic psychedelic settings (Kruger et al., 2025; McNamee et al., 2023) and with the community harm-reduction literature that developed in response (Peluso et al., 2020), and the crisis signs described are of a kind documented at non-trivial rates in large naturalistic samples (Carbonaro et al., 2016; Bouso et al., 2022). The item used here was deliberately broad, covering unexpected touch and uncertain or unremembered consent, so the figure should not be read against a narrower legal or clinical standard.

The study establishes that these events occur among respondents and that an instrument of this kind can capture them; it cannot establish how often they occur, in whom, or under what conditions. The cross-tabulations against participant-side proxies pair a lifetime outcome with single-ceremony exposures, are powered neither to confirm nor to rule out an association, and did not survive correction, so treating any of them as an actionable signal would overstate them considerably. The safety domain justified its inclusion by identifying a measurement priority, not

by producing an estimate.

4.4 The preparation finding as a design lesson

The preparation-to-satisfaction comparison is this study's pre-specified comparison and its most fragile result, and those two facts should be held together rather than traded off. The direction was positive and the effect sizes were not small, but the no-preparation group contained five respondents answering the outcome, the three-level sensitivity analysis was non-significant, and the adjusted model placed preparation short of significance with a confidence interval spanning nearly two orders of magnitude. Nothing here establishes the finding.

The durable contribution is diagnostic: the preparation split was unbalanced enough to dissolve the original four-arm design and remains the binding constraint on the primary analysis, so any successor study will need either a sampling strategy that reaches unprepared participants in usable numbers or a question that does not depend on that contrast. Reporting the deviation rather than absorbing it quietly is what makes the constraint visible to the next design.

4.5 Limitations

The sample was non-probability, unweighted, and self-selected into an online survey; the concentration in higher education categories and favorable economic self-ratings describe who responded, not who participates, and nothing here supports a statement about how common ceremonialentheogen use is in Puerto Rico. Roughly 6% of those answering the residency item lived outside Puerto Rico, further tempering any Puerto Rico-specific reading, and the eligibility criteria were self-assessed rather than enforced.

Role subsamples were small and unevenly sized, and the facilitator branch of eight supports no rate and no association test. Because 7 of those 8 also answered as participants, the facilitator-side and participant-side statistics are not independent samples and should not be read as two separate groups. The reports are nonetheless unlinked at the ceremony level, which is not a minor inconvenience: it eliminated the protocol's integration-proxy comparison, its perceived-safety-by-

training analysis, and its mixed harm-reduction composite, and no statement in this paper connects what facilitators reported doing to what participants reported experiencing.

Two referent mismatches limit the associations reported here: the safety cross-tabulations pair a lifetime outcome with most-recent-ceremony exposures, and the primary comparison pairs a most-recent-ceremony outcome with a general-practice exposure. All measures were single-timepoint self-report, and for most respondents the most recent ceremony was distant—30 of 66 answering reported more than 12 months—so recall error is plausible and unmeasurable. The instrument contains no integration measure, is unvalidated, and produced mixed first reliability evidence (Supplementary Table S9) alongside two documented skip-logic errors (Supplementary Material §S6).

4.6 Conclusion

Within a self-selected sample of 73 adults, ceremonial entheogen use in Puerto Rico was described by a middle-aged, university-educated, overwhelmingly Puerto Rico–resident group whose most recent ceremonies were typically single, multi-hour, outdoor events involving psilocybin mushrooms or ayahuasca, described in strongly positive terms. Preparation was near-universal while pre-ceremony screening was not, and the small facilitator subsample described training and emergency planning as informal rather than credentialed. Five of 67 participants answering reported non-consensual physical contact, and facilitators reported witnessing crisis events at a base too small for any rate to be estimated. These findings describe this sample alone, support no estimate of prevalence and no inference about what any practice produces, and take their principal value from identifying which measures, sampling strategies, and instrument corrections a properly powered successor study would require.

Author contributions

YOOO conceived the study, led protocol development and ethics submission, and supervised the project. YOOO, JCVR, JMHT, JMT, PRF, CR, ANGR, and AIRM contributed to instrument de-

velopment and review. JCVR led data curation, formal analysis, and manuscript drafting. JMHT contributed to methodology and analysis review. JMT, PRF, and CR contributed to interpretation and clinical/community context. ANGR and AIRM contributed to recruitment and data collection. All authors reviewed and approved the final manuscript.

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Competing interests

Several authors are affiliated with organizations that conduct, advocate for, or provide services related to psychedelic and entheogen research and practice, including the Puerto Rico Institute for Psychedelic Science, Medicine & Awareness and the Colectivo Psicodélico de Puerto Rico. These affiliations are disclosed as potential non-financial competing interests. No author received financial support from, or holds a financial interest in, any commercial entity with an interest in the subject matter of this article.

Data availability

The de-identified survey data and the complete analysis code that generates every result reported here will be deposited in a public repository on publication, and the digital object identifier will be recorded here. The data contain no direct identifiers by design. Until deposit, both are available from the corresponding author.

Generative AI use declaration

Large language model tooling (Anthropic Claude) was used to assist with statistical analysis code development and with drafting and editing of this manuscript. All analyses were specified by the authors, all outputs were verified against the source data by the authors, and the authors take full responsibility for the content of this article.

References

- Bethlehem, J. (2010). Selection bias in web surveys. *International Statistical Review* 78(2), 161–188.
- Bird, H. R. and I. Canino (1981). The sociopsychiatry of espiritismo: Findings of a study in psychiatric populations of Puerto Rican and other Hispanic children. *Journal of the American Academy of Child Psychiatry* 20(4), 725–740.
- Bouso, J. C., Ó. Andión, J. J. Sarris, M. Scheidegger, L. F. Tófoli, E. S. Opaleye, V. Schubert, and D. Perkins (2022). Adverse effects of ayahuasca: Results from the Global Ayahuasca Survey. *PLOS Global Public Health* 2(11), e0000438.
- Bouso, J. C., E. J. Pedrero-Pérez, S. Gandy, and M. Á. Alcázar-Córcoles (2016). Measuring the subjective: Revisiting the psychometric properties of three rating scales that assess the acute effects of hallucinogens. *Human Psychopharmacology: Clinical and Experimental* 31(5), 356–372.
- Carbonaro, T. M., M. P. Bradstreet, F. S. Barrett, K. A. MacLean, R. Jesse, M. W. Johnson, and R. R. Griffiths (2016). Survey study of challenging experiences after ingesting psilocybin mushrooms: Acute and enduring positive and negative consequences. *Journal of Psychopharmacology* 30(12), 1268–1278.
- Carvalho, L. C., J. Encantado, H. Kettner, C. Timmermann, D. Veiga, and P. J. Teixeira (2025).

A scoping review of research in naturalistic studies with psychedelics. *Journal of Psychoactive Drugs*, 1–14. Online First; volume/issue not yet assigned as of 2026-08-13.

Comas-Díaz, L. (1981). Puerto Rican espiritismo and psychotherapy. *American Journal of Orthopsychiatry* 51(4), 636–645.

Doelter Morales, F. M. and J. E. Martínez Miranda (2020). Hongos mágicos...más allá de las alucinaciones: Análisis político-social y estudio exploratorio del conocimiento y las percepciones que tiene la comunidad de la Universidad de Puerto Rico, Recinto de Río Piedras sobre el uso de los hongos psicodélicos a diciembre de 2020. M.a. thesis, University of Puerto Rico, Río Piedras Campus. Unpublished; repository record <https://repositorio.upr.edu/handle/11721/2540>. No DOI assigned.

Dutton, C., E. North, Y. Chun Tie, J. Oliva, and P. Skeffington (2025). Harm reduction practises for users of psychedelic drugs: A scoping review. *Harm Reduction Journal* 22(1), 158.

Golden, T. L., S. Magsamen, C. C. Sandu, S. Lin, G. M. Roebuck, K. M. Shi, and F. S. Barrett (2022). Effects of setting on psychedelic experiences, therapies, and outcomes: A rapid scoping review of the literature. In *Disruptive Psychopharmacology*, Volume 56 of *Current Topics in Behavioral Neurosciences*, pp. 35–70. Cham: Springer. CrossRef types this work as a book chapter in the *Current Topics in Behavioral Neurosciences* book series; given names and volume/pagination follow the publisher-deposited record.

Gorman, I., E. M. Nielson, A. Molinar, K. Cassidy, and J. Sabbagh (2021). Psychedelic harm reduction and integration: A transtheoretical model for clinical practice. *Frontiers in Psychology* 12, 645246.

Hartogsohn, I. (2017). Constructing drug effects: A history of set and setting. *Drug Science, Policy and Law* 3, 2050324516683325. Article number in place of page range, per the journal's continuous-numbering convention.

- Hughes, M. E. and A. Garcia-Romeu (2024). Ethnoracial inclusion in clinical trials of psychedelics: A systematic review. *eClinicalMedicine* 74, 102711.
- Kopra, E. I., J. A. Ferris, A. R. Winstock, K. P. C. Kuypers, A. H. Young, and J. J. Rucker (2023). Investigation of self-treatment with lysergic acid diethylamide and psilocybin mushrooms: Findings from the Global Drug Survey 2020. *Journal of Psychopharmacology* 37(7), 733–748.
- Koss, J. D. (1980). The therapist-spiritist training project in Puerto Rico: An experiment to relate the traditional healing system to the public health system. *Social Science & Medicine. Part B: Medical Anthropology* 14(4), 255–266.
- Kruger, D. J., J. S. Aday, C. W. Fields, N. Kolbman, N. Glynos, J. Barron, M. Herberholz, and K. F. Boehnke (2025). Psychedelic therapist sexual misconduct and other adverse experiences among a sample of naturalistic psychedelic users. *Psychedelic Medicine* 3(1), 41–47.
- Kruger, D. J., J. Barron, M. Herberholz, and K. F. Boehnke (2023). Preferences and support for psychedelic policies and practices among those using psychedelics. *Journal of Psychoactive Drugs* 55(5), 650–659.
- Legislative Assembly of Puerto Rico (1971). Ley de Sustancias Controladas de Puerto Rico. Ley Núm. 4 de 23 de junio de 1971, según enmendada; codified at 24 L.P.R.A. § 2101 et seq. Puerto Rico’s territorial controlled-substances statute, aligned with the federal Controlled Substances Act.
- Marcus, O. and B. Rush (2026). Psychedelics for the treatment of substance use disorders: A narrative review of the literature. *Substance Use & Addiction Journal* 47(1), 279–296. Published online first 2025-07-27; cited here to the 2026 print issue, 47(1):279–296.
- McNamee, S., N. Devenot, and M. Buisson (2023). Studying harms is key to improving psychedelic-assisted therapy—participants call for changes to research landscape. *JAMA Psychiatry* 80(5), 411–412. CrossRef records the first page only (411).

- Nayak, S. M., H. Jackson, N. D. Sepeda, D. S. Mathai, S. So, A. Yaffe, H. Zaki, T. J. Brasher, M. X. Lowe, D. R. P. Jolly, F. S. Barrett, R. R. Griffiths, J. C. Strickland, M. W. Johnson, and A. Garcia-Romeu (2023). Naturalistic psilocybin use is associated with persisting improvements in mental health and wellbeing: Results from a prospective, longitudinal survey. *Frontiers in Psychiatry* 14, 1199642.
- Pagni, B. A., A. Halman, J. Sarris, R. Chenhall, M. P. Bogenschutz, and D. Perkins (2025). Long-term mental health and wellbeing outcomes associated with naturalistic ayahuasca consumption. *Journal of Psychoactive Drugs*, 1–11. Online First; volume/issue not yet assigned as of 2026-08-13.
- Peluso, D., E. Sinclair, B. Labate, and C. Cavnar (2020). Reflections on crafting an ayahuasca community guide for the awareness of sexual abuse. *Journal of Psychedelic Studies* 4(1), 24–33.
- Pilecki, B., J. B. Luoma, G. J. Bathje, J. Rhea, and V. F. Narloch (2021). Ethical and legal issues in psychedelic harm reduction and integration therapy. *Harm Reduction Journal* 18(1), 40.
- Robinson, J., N. Wallace, E. Cassidy, S. Lake, and P. Lucas (2026). Field notes from inner space: Global psychedelic survey participants report health benefits, personal growth, and life transformations. *Contemporary Drug Problems*. Online First; volume/issue/pages not yet assigned as of 2026-08-13.
- Ruffell, S. G. D., N. Netzband, W. Tsang, M. Davies, M. Butler, J. J. H. Rucker, L. F. Tófoli, E. L. Dempster, A. H. Young, and C. J. A. Morgan (2021). Ceremonial ayahuasca in amazonian retreats—mental health and epigenetic outcomes from a six-month naturalistic study. *Frontiers in Psychiatry* 12, 687615.
- Siegel, J. S., J. E. Daily, D. A. Perry, and G. E. Nicol (2023). Psychedelic drug legislative reform and legalization in the US. *JAMA Psychiatry* 80(1), 77–83.

United States Congress (1970). Controlled substances act. 21 U.S.C. § 812, Pub. L. No. 91-513, 84 Stat. 1242. Codifies the federal Schedule I classification of psilocybin, DMT, and other classic psychedelics that applies uniformly across U.S. states and territories, including Puerto Rico.

Vélez Rodríguez, J. C., Y. Duquesne-Maldonado, J. M. Hernández Torres, I. Da'Luz Sant'Ana, and C. M. Nazario (2026). Psilocybin use in Puerto Rico: Patterns, motivations, and personality correlates from an online survey. *Journal of Psychedelic Studies*. Online First, published 2026-04-15; volume/issue/pages not yet assigned as of 2026-08-13. Companion output from the same research team as the present study; distinct in scope (single-substance psilocybin use-pattern and personality survey vs. this study's multi-substance ceremonial-practice and safety focus).

Zerrate, M. C., S. B. VanBronkhorst, J. Klotz, A. A. Caraballo, G. Canino, H. R. Bird, and C. S. Duarte (2022). Espiritismo and santería: A gateway to child mental health services among Puerto Rican families? *Child and Adolescent Psychiatry and Mental Health* 16(1), 3.

Tables and Figures

Table 1: Sociodemographic characteristics of the analytic sample ($N = 73$). Bases differ by item and are stated with each; counts are exact and percentages are of respondents answering.

Characteristic	Category	<i>n</i>	% answering
Age, years ($n = 72$)	Mean 42.4 (SD 11.9); median 41 (IQR 34–50); range 21–71		
Gender ($n = 71$)	Woman	35	49
	Man	30	42
	Non-binary	4	6
	Prefer not to answer	2	3
Education ($n = 72$)	Bachelor's degree	34	47
	Master's degree	19	26
	Associate's degree	7	10
	High school diploma / GED	6	8
	Doctorate	3	4
	Prefer not to answer	3	4
Economic situation ($n = 72$)	Comfortable	44	61
	Tight	20	28
	Difficult	6	8
	Prefer not to answer	2	3
Country of residence ($n = 72$)	Puerto Rico	68	94
	United States	3	4
	Spain	1	1

Table 2: Most-recent-ceremony context, participant-reported. Participant branch only: 72 respondents identified as participants, of whom 3 were not routed to these items by the continuation gate described in Methods and appear as structural skips. Item nonresponse is the residual between the per-item base and 69. Counts are exact; percentages are of respondents answering the item.

Item	Category	<i>n</i>	% answering
Time since last ceremony (<i>n</i> = 66 answering)	More than 12 months ago	30	45
	2 to 6 months ago	20	30
	7 to 12 months ago	11	17
	During the past month	4	6
	Prefer not to answer	1	2
	<i>Structural skip (not routed)</i>	3	—
	<i>Item nonresponse</i>	3	—
Place (<i>n</i> = 67 answering)	Natural setting (forest, beach, mountain)	37	55
	Private home	15	22
	Retreat center or wellness facility	9	13
	Other	3	4
	Urban or residential space used informally	2	3
	Prefer not to answer	1	1
	<i>Structural skip (not routed)</i>	3	—
	<i>Item nonresponse</i>	2	—
Format (<i>n</i> = 65 answering)	Single ceremony	48	74
	Retreat or series of ceremonies	13	20
	Prefer not to answer	3	5
	Not sure	1	2
	<i>Structural skip (not routed)</i>	3	—
	<i>Item nonresponse</i>	4	—
Duration (<i>n</i> = 65 answering)	4 to 6 hours	19	29
	Overnight (~12+ hours)	19	29
	7 to 12 hours	14	22
	1 to 3 hours	7	11
	2 to 3 days	4	6
	4 to 6 days	1	2
	Less than 1 hour	1	2
	<i>Structural skip (not routed)</i>	3	—
	<i>Item nonresponse</i>	4	—

Table 3: Participant-reported protective practices and safety outcomes. Bases are per item and differ because each is asked under its own skip condition; the 3 structural skips are the dual-role respondents not routed to the participant section. Counts are exact throughout. The two non-consensual contact items differ in referent: the first asks about any ceremony ever, the second about the most recent ceremony only.

Item	Category	<i>n</i>	% answering
Self-directed preparation (<i>n</i> = 67)	Yes	55	82
	Sometimes	6	9
	No	6	9
	<i>Structural skip / item nonresponse</i>	3 / 2	—
Knew facilitator’s background (<i>n</i> = 67)	Yes	54	81
	A little	8	12
	No	4	6
	Prefer not to answer	1	1
	<i>Structural skip / item nonresponse</i>	3 / 2	—
Facilitator asked screening questions (<i>n</i> = 66)	Yes	39	59
	No	15	23
	Not sure	7	11
	Prefer not to answer	5	8
	<i>Structural skip / item nonresponse</i>	3 / 3	—
Spoke to a health professional beforehand (<i>n</i> = 65)	No	54	83
	Yes	8	12
	Prefer not to answer	2	3
	Not sure	1	2
	<i>Structural skip / item nonresponse</i>	3 / 4	—
Non-consensual physical contact, ever (<i>n</i> = 67)	No	61	91
	Yes	5	7

Item	Category	<i>n</i>	% answering
	Not sure	1	1
	<i>Structural skip / item nonresponse</i>	3 / 2	—
Non-consensual physical contact, most recent ceremony (<i>n</i> = 65)	No	62	95
	Yes	2	3
	Prefer not to answer	1	2
	<i>Structural skip / item nonresponse</i>	3 / 4	—
Harm-reduction practice index (<i>n</i> = 54 complete cases)	All three practices	35	65
	Two practices	16	30
	One practice	3	6
	No practices	0	0

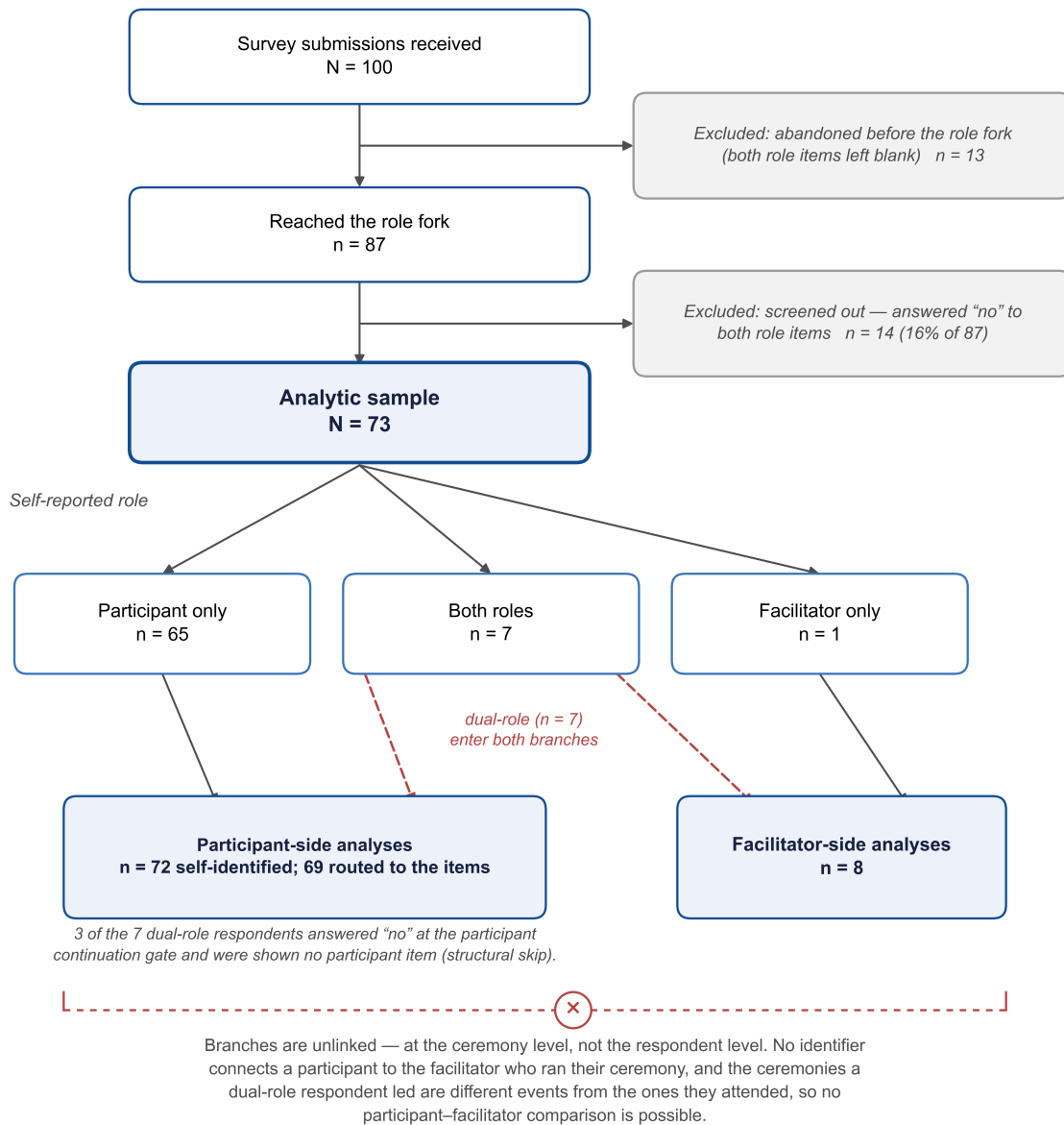
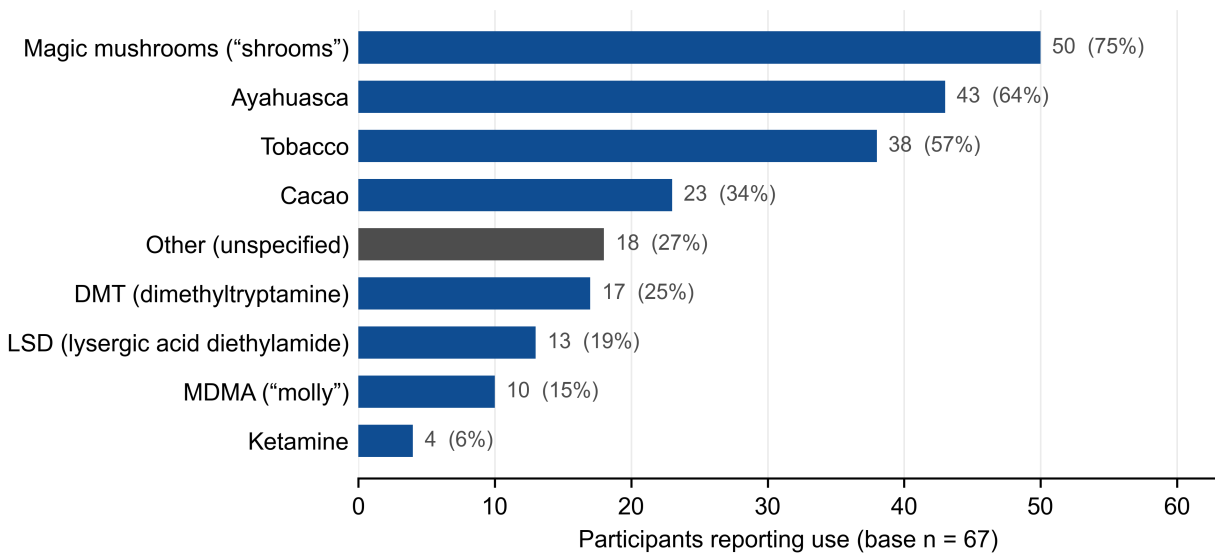
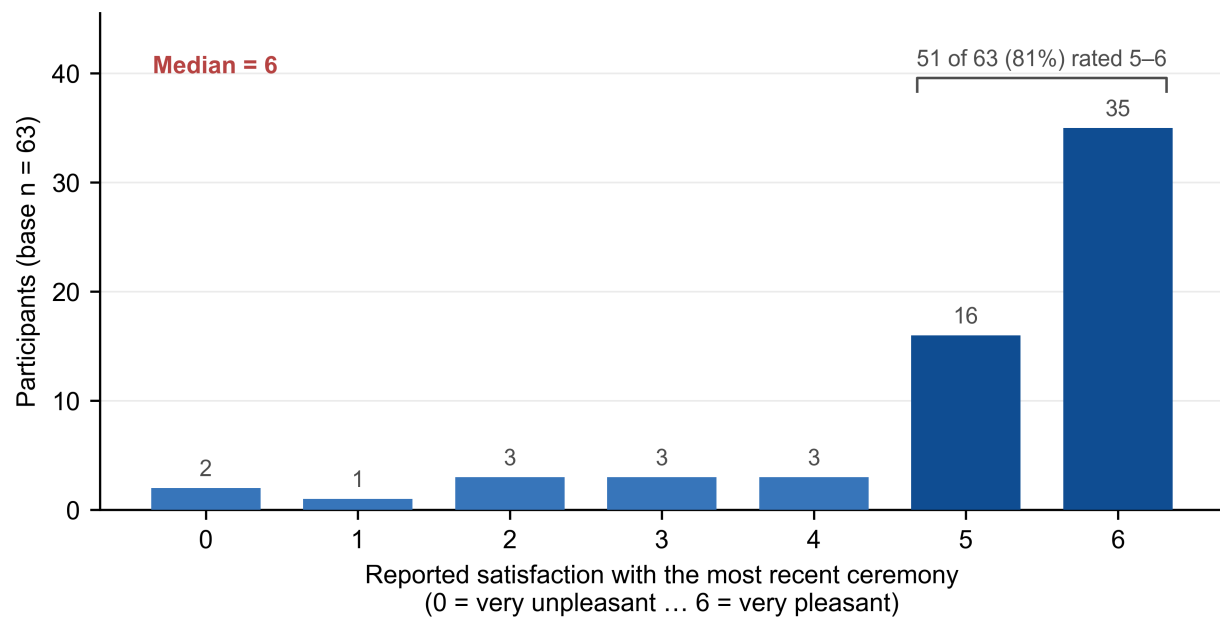


Figure 1: Sample construction and role structure. Of 100 submissions, 13 were abandoned before the role fork and 14 of the remaining 87 screened out by answering “no” to both role items, leaving an analytic sample of $N = 73$. Roles are not mutually exclusive: the 7 dual-role respondents enter both branches, giving 72 participant-side and 8 facilitator-side reports, and 3 of those 7 were stopped by the participant continuation gate, so 69 respondents were actually routed to the participant items. The branches are unlinked at the ceremony level — no identifier connects a participant to the facilitator who ran their ceremony, and the ceremonies a dual-role respondent led are different events from those they attended — so no participant–facilitator comparison is possible.



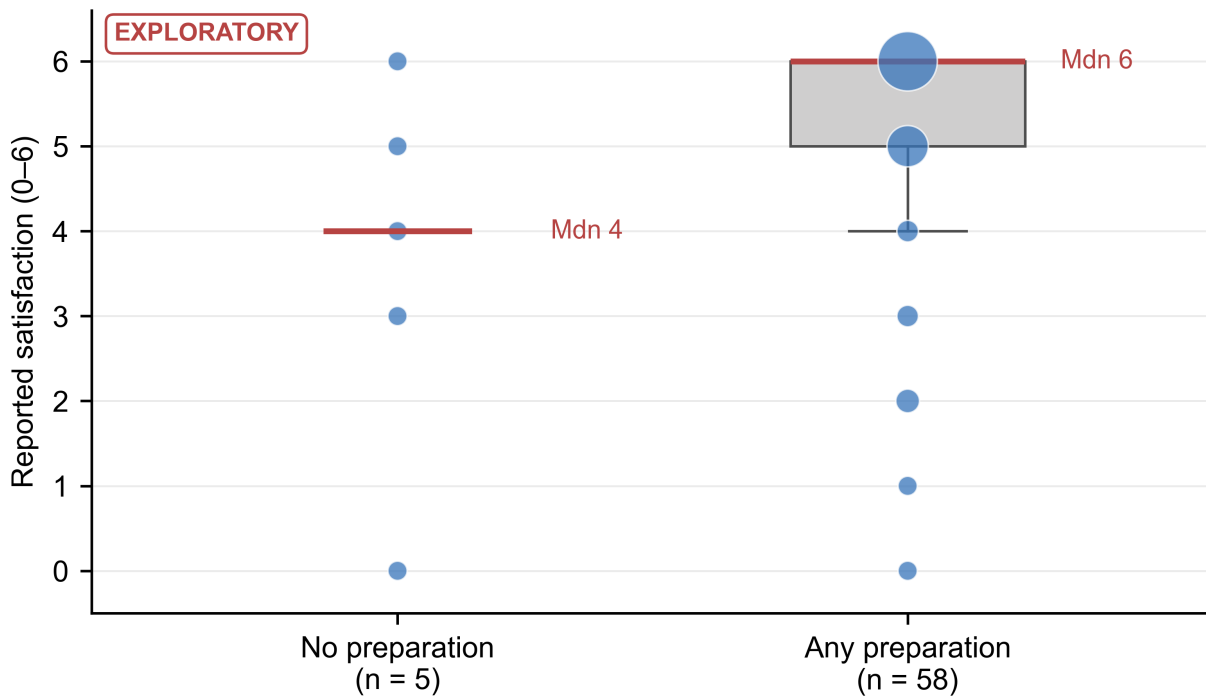
*Select-multiple item: percentages are of the base and do not sum to 100%.
 "Other" is shown in grey because its free-text follow-up was completed by only 3 respondents, so the category cannot be interpreted.*

Figure 2: Substances used across all ceremonies attended, participant-reported (select-multiple; base $n = 67$ participants answering). Bars are labeled with the count and the percentage of that base; because respondents could select more than one substance, percentages need not sum to 100%. The "other" bar is drawn in grey because its free-text follow-up was completed by only 3 respondents, leaving the category uninterpretable. Distinct from the facilitator-led substance variable and from the most-recent-ceremony substance variable.



Of the 72 respondents who identified as participants, 63 gave a numeric score; 1 chose “prefer not to answer”, 3 were never routed to the participant section (structural skip), and 5 did not answer.

Figure 3: Distribution of reported satisfaction with the most recent ceremony (0 = very unpleasant to 6 = very pleasant). Of the 72 respondents identifying as participants, 63 gave a numeric score and form the base; the remaining 9 comprise 1 “prefer not to answer”, 3 structural skips at the continuation gate, and 5 item nonresponses. Median 6, with 51 of 63 (81%) answering 5 or 6 — the ceiling that limits how much variance any predictor can explain.



Marker area is proportional to the number of respondents giving that score. The no-preparation group ($n = 5$) is drawn as its raw points and median tick only: a box would imply distributional precision five observations cannot support.

Figure 4: Reported satisfaction by self-directed preparation status. Any preparation, $n = 58$ answering the outcome (median 6, IQR 5–6); no preparation, $n = 5$ (median 4, IQR 3–5). Marker area is proportional to the number of respondents giving each score, since scores tie heavily at the ceiling. The no-preparation group is drawn as its five raw points and a median tick rather than a box, which at that n would imply distributional precision the data cannot support. **Exploratory:** this comparison is suggestive, not confirmatory.